ಜಿ. ಎಂ. ವಿಶ್ವವಿದ್ಯಾಲಯ
GM UNIVERSITY
P. B. Road, Davanagere – 577 006 KARNATAKA | INDIA

Introduction to Bayesian Learning

Bayesian Learning is a probability-based machine learning approach that uses Bayes' Theorem to make predictions and decisions. It helps in updating the probability of a hypothesis whenever new evidence or information is available.

In simple words, Bayesian learning learns from past data and improves predictions when new data is received.

It is based on the idea:

"Initial belief + New evidence = Updated belief"

Bayesian learning calculates probabilities for different outcomes and selects the outcome with the highest probability.

Bayes' Theorem

Bayes' Theorem is a probability theorem used to find the probability of an event after getting new information or evidence. It helps in updating the existing probability based on additional knowledge.

In simple words, Bayes' theorem tells us how likely something is when some related information is already known.

It is one of the most important concepts in probability-based learning and machine learning.

The formula for Bayes' theorem is:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

- $P(A|B)$ = Probability of event A given that event B has occurred (Posterior Probability)
- $P(B|A)$ = Probability of event B given that event A has occurred (Likelihood)
- $P(A)$ = Initial probability of event A (Prior Probability)
- $P(B)$ = Total probability of event B (Evidence)

Bayes' theorem works by combining prior knowledge with new evidence to produce a better and updated probability.

Example

Suppose a boy usually participates in volleyball tournaments. If we know that the boy is sick, Bayes' theorem helps us calculate the probability that he will still participate in the tournament.

Thus, Bayes' theorem is very useful in decision-making and prediction problems.

Applications of Bayes' Theorem

- Spam email filtering
- Medical diagnosis
- Weather forecasting
- Machine learning algorithms
- Recommendation systems

Conclusion

Bayes' theorem is an important statistical method used to calculate conditional probabilities. It helps in improving predictions by using prior information and new evidence together.

Term	**Formula**	**Simple Meaning**
Posterior Probability	(P(A|B))	Final updated probability
Likelihood	(P(B|A))	Probability of evidence if event is true
Prior Probability	(P(A))	Initial probability before evidence
Evidence	(P(B))	Total probability of evidence

Very Easy Memory Trick

Prior

☞ What we believed BEFORE.

Evidence

☞ New information we got.

Likelihood

☞ How strongly the evidence supports the event.

Posterior

☞ Final updated belief AFTER using evidence.

Consider a boy who has a volleyball tournament on the next day, but today he feels sick. It is unusual that there is only a 40% chance he would fall sick since he is a healthy boy. Now, find the probability of the boy participating in the tournament. The boy is very much interested in volleyball, so there is a 90% probability that he would participate in tournaments and 20% that he will fall sick given that he participates in the tournament.

Bayes' theorem formula:

$$P(T|S) = \frac{P(S|T) \times P(T)}{P(S)}$$

Where:

- T = Boy participates in tournament
- S = Boy is sick

1. Prior Probability

$$P(T) = 0.9$$

Meaning

☞ Probability of the boy participating **before knowing he is sick**.

In Our Example

The boy loves volleyball.

So initially:

☞ There is a 90% chance he participates.

This is called:

✅ Prior Probability

Because it is our belief BEFORE new information.

2. Evidence

$$P(S) = 0.4$$

Meaning

☞ Probability that the boy is sick.

In Our Example

We know:

☞ There is a 40% chance the boy becomes sick.

This sickness information is the evidence.

So:

✅ Evidence = Probability of sickness

3. Likelihood

$$P(S|T) = 0.2$$

Meaning

☞ Probability the boy is sick GIVEN that he participates.

In Our Example

Suppose the boy still goes to tournaments even sometimes when sick.

So:

☞ Among participation situations, sickness happens 20% of the time.

This tells how strongly sickness and participation are related.

So this is called:

✅ Likelihood

4. Posterior Probability

$$P(T|S)$$

Meaning

☞ Probability the boy participates AFTER knowing he is sick.

This is the final answer we calculate using Bayes theorem.

After calculation:

$$P(T|S) = \frac{0.2 \times 0.9}{0.4}$$

$$P(T|S) = 45\%$$

Meaning:

☞ Even if the boy is sick, there is still a 45% chance he participates.

This updated probability is called:

✅ Posterior Probability

Probability the boy participates **given that he is sick**

$$P(T|S)$$

Meaning:

☞ “If we already know he is sick, what is the chance he still plays?”

Use Bayes’ Theorem

$$P(T|S) = \frac{P(S|T) \times P(T)}{P(S)}$$

Now put the values:

$$P(T|S) = \frac{0.2 \times 0.9}{0.4}$$

Final Answer

$$P(T|S) = 0.45 = 45\%$$

So there is a **45% chance** the boy will still participate in the tournament even though he is sick.

Naïve Bayes Theorem

Naïve Bayes is a supervised machine learning algorithm used for classification problems. It is based on Bayes’ Theorem and is mainly used to predict the class or category of data.

It is called “Naïve” because it assumes that all attributes are independent of each other.

For example, while predicting whether a student gets a job offer, Naïve Bayes assumes that:

- CGPA,
- Communication Skills,
- Practical Knowledge, and

- Interactiveness

Definition

Naïve Bayes is a probabilistic classification algorithm that uses Bayes' theorem with the assumption that features are conditionally independent.

Naïve Bayes Formula

In Naïve Bayes, multiple attributes are considered:

$$P(C|X) = P(x_1|C) \times P(x_2|C) \times P(x_3|C) \times \cdots \times P(C)$$

Where:

- C = Class label (Yes/No)
- X = Input attributes
- x_1, x_2, x_3 = Features

Q. Assess a student's performance using Naïve Bayes algorithm with the dataset provided in Table 8.1. Predict whether a student gets a job offer or not in his final year of the course.

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
≥9	Yes	Very Good	Good	Yes
≥8	No	Good	Moderate	Yes
≥9	No	Average	Poor	No
<8	No	Average	Good	No
≥8	Yes	Good	Moderate	Yes
≥9	Yes	Good	Moderate	Yes
<8	Yes	Good	Poor	No
≥9	No	Very Good	Good	Yes

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
≥8	Yes	Good	Good	Yes
≥8	Yes	Average	Good	Yes

Working of Naïve Bayes

Step 1

Collect training dataset.

Example:

Student data with CGPA, communication skills, etc.

Step 2

Calculate prior probabilities.

Example:

Probability of Yes

$$P(Yes) = \frac{7}{10} = 0.7$$

Probability of No

$$P(No) = \frac{3}{10} = 0.3$$

Step 3

Calculate likelihood probabilities for each attribute.

Calculate Likelihoods for YES

We calculate probabilities using only rows where Job Offer = Yes.

Total Yes rows = 7

1. CGPA ≥8 given Yes

All 7 Yes rows have CGPA ≥8.

$$P(CGPA \geq 8|Yes) = \frac{7}{7} = 1$$

2. Interactiveness = Yes given Yes

"Yes" appears 5 times.

$$P(Interactive = Yes|Yes) = \frac{5}{7}$$

3. Practical Knowledge = Good given Yes

"Good" appears 4 times.

$$P(Practical = Good|Yes) = \frac{4}{7}$$

4. Communication Skills = Good given Yes

"Good" appears 4 times.

$$P(Communication = Good|Yes) = \frac{4}{7}$$

Step 5: Calculate Posterior Probability for YES

Using Naïve Bayes:

$$P(Yes|X) = P(CGPA|Yes) \times P(Interactive|Yes) \times P(Practical|Yes) \times P(Communication|Yes) \times P(Yes)$$

Substitute values:

$$P(Yes|X) = 1 \times \frac{5}{7} \times \frac{4}{7} \times \frac{4}{7} \times 0.7$$

$$= 0.163$$

Step 6: Calculate Likelihoods for NO

Total No rows = 3

1. CGPA ≥8 given No

Appears 1 time.

$$P(CGPA \geq 8|No) = \frac{1}{3}$$

2. Interactiveness = Yes given No

Appears 1 time.

$$P(Interactive = Yes|No) = \frac{1}{3}$$

3. Practical Knowledge = Good given No

Appears 1 time.

$$P(Practical = Good|No) = \frac{1}{3}$$

4. Communication Skills = Good given No

Appears 1 time.

$$P(Communication = Good|No) = \frac{1}{3}$$

Step 7: Calculate Posterior Probability for NO

$$P(No|X) = P(CGPA|No) \times P(Interactive|No) \times P(Practical|No) \times P(Communication|No) \times P(No)$$

Substitute values:

$$P(No|X) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times 0.3$$

$$= 0.0037$$

Step 8: Compare Probabilities

Class	Probability
Yes	0.163
No	0.0037

Since:

0.163>0.0037

Final Prediction

✅ **Job Offer = YES**

The student with following attributes is likely to get a job offer.

Attribute	Value
CGPA	≥8
Interactiveness	Yes
Practical Knowledge	Good
Communication Skills	Good